HITEC University Taxila

Department of Cyber Security

Course: Essentials of Artificial Intelligence

Lab Manual 4

Submitted By : Muhammad Sannan Azmat

Reg No: 23-CyS-010

Submitted To: Mubashir Iqbal

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Lab Tasks

Task 1. Write a Python class named Circle constructed by a radius and two methods which will compute the area and the perimeter of a circle.

Input

```
lab4 Task 1.py > Circle > perimeter
import math

class Circle:
    def __init__(self, radius):
        self.radius = radius

    def area(self):
        return math.pi * (self.radius ** 2)

    def perimeter(self):
        return 2 * math.pi * self.radius
c = Circle(5)
print("Area:", c.area())
print("Perimeter:", c.perimeter())
```

Output

```
PS S:\AI Lab> & C:/Users/HP/AppData/Local/Pr
Task 1.py"
Area: 78.53981633974483
Perimeter: 31.41592653589793
PS S:\AI Lab>
```

Task 2. Write a Python class to implement pow(x, n). Do not use built-in math.pow() function for this code. Also handle this code for all +ve as well as -ve numbers.

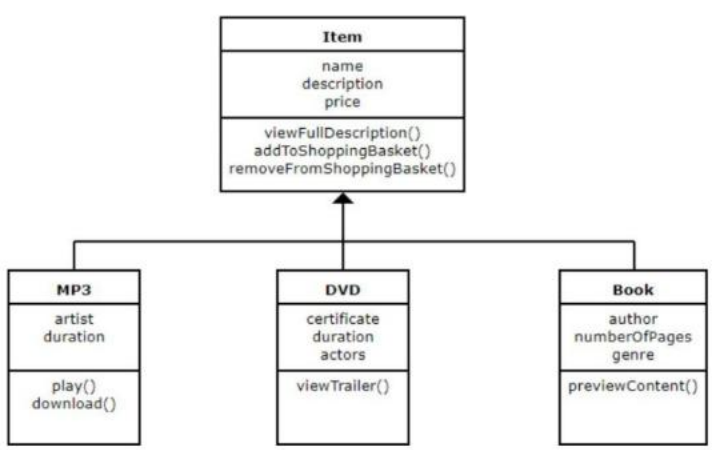
Input

```
lab4 Task 2.py > Power > __init__
class Power:
    def __init__(self, x, n):
        self.x = x
        self.n = n
    def compute(self):
        if self.n == 0:
            return 1
        if self.n > 0:
            result = 1
            for _ in range(self.n):
                result *= self.x
            return result
        else:
            result = 1
            for _ in range(-self.n):
                result *= self.x
            return 1 / result
p1 = Power(2, 3)   # 2^3 = 8
print("2^3 =", p1.compute())
p2 = Power(2, -3)  # 2^-3 = 1/8
print("2^-3 =", p2.compute())
p3 = Power(5, 0)   # 5^0 = 1
print("5^0 =", p3.compute())
```

Output

```
Task 2.py
2^3 = 8
2^-3 = 0.125
5^0 = 1
PS S:\AI Lab>
```

Task 3. Write a Python code for the following figure



Input

```python
class Item:
    def __init__(self, name, description, price):
        self.name = name
        self.description = description
        self.price = price
    def viewFullDescription(self):
        return f"{self.name}: {self.description} - ${self.price:.2f}"
    def addToShoppingBasket(self, basket):
        basket.append(self)
        print(f"{self.name} added to basket.")
    def removeFromShoppingBasket(self, basket):
        if self in basket:
            basket.remove(self)
            print(f"{self.name} removed from basket.")
        else:
            print(f"{self.name} not found in basket.")
class MP3(Item):
    def __init__(self, name, description, price, artist, duration):
        super().__init__(name, description, price)
        self.artist = artist
        self.duration = duration
    def play(self):
        print(f"Playing '{self.name}' by {self.artist} ({self.duration})...")
    def download(self):
        print(f"Downloading '{self.name}'...")
```

```python
lass DVD(Item):
    def __init__(self, name, description, price, certificate, duration, actors):
        super().__init__(name, description, price)
        self.certificate = certificate
        self.duration = duration
        self.actors = actors
    def viewTrailer(self):
        print(f"Viewing trailer for '{self.name}' ({self.duration}, {self.certificate})...")
lass Book(Item):
    def __init__(self, name, description, price, author, numberOfPages, genre):
        super().__init__(name, description, price)
        self.author = author
        self.numberOfPages = numberOfPages
        self.genre = genre
    def previewContent(self):
        print(f"Previewing '{self.name}' by {self.author} ({self.genre}, {self.numberOfPages
f __name__ == "__main__":
    basket = []
    song = MP3("Song A", "Hit single", 1.99, "Artist X", "3:45")
    movie = DVD("Movie B", "Action film", 9.99, "PG-13", "120 min", ["Actor Y", "Actor Z"])
    novel = Book("Book C", "Fantasy novel", 14.99, "Author W", 350, "Fantasy")
    song.play()
    song.download()
    song.addToShoppingBasket(basket)
    movie.viewTrailer()
```

```python
    movie.addToShoppingBasket(basket)
    novel.previewContent()
    novel.addToShoppingBasket(basket)
    print("\n--- Shopping Basket ---")
    for item in basket:
        print(item.viewFullDescription())
    song.removeFromShoppingBasket(basket)
    print("\n--- Updated Basket ---")
    for item in basket:
        print(item.viewFullDescription())
```

Output

```
task 3.py"
Playing 'Song A' by Artist X (3:45)...
Downloading 'Song A'...
Song A added to basket.
Viewing trailer for 'Movie B' (120 min, PG-13)...
Movie B added to basket.
Previewing 'Book C' by Author W (Fantasy, 350 pages)...
Book C added to basket.

--- Shopping Basket ---
Song A: Hit single - $1.99
Movie B: Action film - $9.99
Book C: Fantasy novel - $14.99
Song A removed from basket.

Book C: Fantasy novel - $14.99
Song A removed from basket.

Song A removed from basket.


--- Updated Basket ---
Movie B: Action film - $9.99
Book C: Fantasy novel - $14.99
```

Conclusion

This lab reinforced key concepts of object-oriented programming in Python. The Circle class demonstrated encapsulation and method-based computation, while the custom pow(x, n) function highlighted algorithmic problem-solving without built-in libraries. The shopping system hierarchy (Item, MP3, DVD, Book) illustrated inheritance, polymorphism, and specialization, showing how shared attributes can be extended with unique behaviors. Overall, the exercises strengthened practical skills in designing modular, reusable, and maintainable code.